Proton in a capacitor

X16 (2016) — English translation

In the middle of a flat capacitor with a distance between the plates $d = 2$ cm there is a proton. Plate No. 1 of the capacitor is grounded, and at some point in time, alternating voltage $U(t) = U_0 \sin \omega t$, where $U_0 = 1$ мВ, $\omega = 10^5$ с$^{-1}$, is applied to plate No. 2. After some time τ the proton collides with one of the plates.

A1	Which (1st or 2nd) plate did the proton collide with? Rate the time τ.	**3.00**

Ignore the force of gravity and the field of induction charges. Proton charge $e = 1.6 \cdot 10^{-19}$ Кл, mass $m = 1.67 \cdot 10^{-27}$ kg.

Source: https://pho.rs/p/3999